

Between-Subject Variability in Memory Modulation

Patient-level linear models using the behavioral first-session summary file

Requested outcome: avg_dprime_diff. The behavioral file does not contain a column with that exact name; the available patient-level outcome is avg_stim_dprime_diff, which was used here.

Critical modeling point: this file has one row per patient (54 rows, 54 unique patients). Because the outcome is already collapsed to a single number per person, a subject random intercept is not identifiable. The correct analysis at this level is a patient-level linear model, not an MLM with (1 | subject).

Executive Summary

- Unadjusted sex difference: males $M = 0.179$, females $M = -0.039$, mean difference = 0.218, Welch $t(42.19) = 2.781$, $p = .008$.
- Primary adjusted model: $R^2 = 0.313$, adjusted $R^2 = 0.231$, $F(5, 42) = 3.823$, $p = .006$.
- Sequential build: sex entered first (model $R^2 = 0.183$, step $p = .002$); stim trajectory entered second (delta $R^2 = 0.071$, step $p = .044$).
- IED_freq and Memory_Z did not improve model fit in the smaller subsets where they were available.

Interpretive frame

These models do not explain the original trial-level ICC directly. Instead, they test whether the requested patient-level covariates are associated with between-subject differences in the already-collapsed memory modulation score.

Data Availability and Descriptives

Table 1
Variable availability in the behavioral summary file

Variable	Label	Non-missing	Missing	Percent available
avg_stim_dprim	Memory modulation (avg_stim_dprime_diff)	54	0	100.0%
sex	Sex (male vs female)	54	0	100.0%
stim_trajectory	Stim trajectory (STG vs MTG; frontal excluded)	48	6	88.9%
age	Age (years)	54	0	100.0%
stim_hemispher	Stim hemisphere (R vs L)	54	0	100.0%
stim_DB	Stim amplitude (DB)	54	0	100.0%
IED_freq	IED frequency (Likert)	44	10	81.5%
Memory_Z	Pre-surgical baseline memory (Memory_Z)	31	23	57.4%

Note. Stim trajectory is available for 48/54 patients after excluding the 3 frontal-trajectory cases from modeling.
Memory_Z is available for 31/54 patients and is concentrated in the Original, Timing, and Duration cohorts.
IED_freq is available for 44/54 patients.

In the raw patient-level outcome, males showed higher avg_stim_dprime_diff than females. That unadjusted difference is useful descriptively, but the primary question is whether it survives after accounting for stimulation trajectory, age, stimulation hemisphere, and stimulation amplitude.

Table 2
Outcome descriptives by sex

Group	N	Mean	SD	Median	Min	Max
female	34	-0.039	0.290	-0.053	-0.526	0.840
male	20	0.179	0.271	0.230	-0.779	0.440

Note. Means are computed on avg_stim_dprime_diff. Positive values indicate greater memory performance on stimulated relative to non-stimulated trials.

Memory_Z subset composition: Original (n=14), Timing (n=12), Duration (n=5). Within that subset, stim_DB has no variation: all available cases are 0.5 DB. Stim trajectory is modeled as STG versus MTG, with frontal trajectories excluded to match the separate trajectory test.

Primary Full-Sample Model

Primary patient-level model using the full cohort. This is the cleanest adjusted test because sex, age, stimulation hemisphere, and stimulation amplitude are available broadly, and stimulation trajectory is available for the non-frontal MTG/STG subset.

Table 3
Primary model fit summary

N	Predictors	R2	Adj R2	F(df)	Model p
48	5	0.313	0.231	3.823 (5, 42)	.006

~~Note. The primary model includes sex first, then stim trajectory, age, stimulation hemisphere, and stimulation amplitude.~~

Table 4
Primary model coefficients

Term	B	SE	t	p	95% CI
Intercept	0.087	0.184	0.470	.640	[-0.285, 0.458]
Sex (male vs female)	0.152	0.084	1.794	.080	[-0.019, 0.322]
Stim trajectory (STG vs MTG; frontal	0.274	0.122	2.249	.030	[0.028, 0.521]
Age (years)	0.002	0.004	0.434	.666	[-0.006, 0.009]
Stim hemisphere (R vs L)	-0.066	0.079	-0.836	.408	[-0.224, 0.093]
Stim amplitude (DB)	-0.270	0.190	-1.424	.162	[-0.652, 0.113]

~~Note. Female, MTG trajectory, and left hemisphere are the reference groups for binary predictors.~~
The primary adjusted model was significant overall, $F(5, 42) = 3.823$, $p = .006$, and accounted for 0.313 of the variance in avg_stim_dprime_diff. In the requested sequence, sex entered first and produced model $R^2 = 0.183$ (step $p = .002$); adding stim trajectory next increased R^2 by 0.071 (step $p = .044$). The adjusted coefficient for sex was $B = 0.152$ (95% CI [-0.019, 0.322]), $t = 1.794$, $p = .080$, and the adjusted coefficient for STG versus MTG trajectory was $B = 0.274$ (95% CI [0.028, 0.521]), $t = 2.249$, $p = .030$.

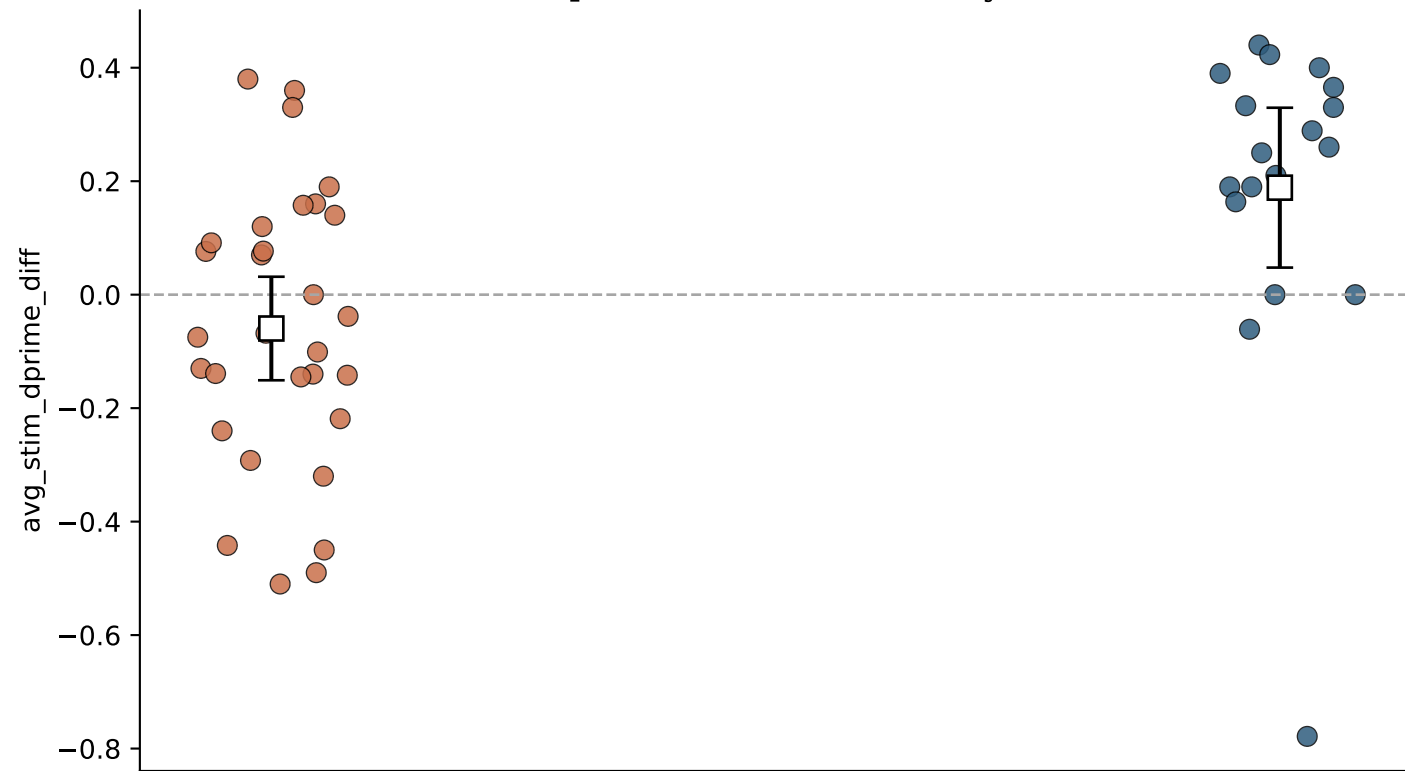
Table 5
Sequential model building in the requested order

Step	Added predictor	Model R2	Delta R2	F change	p
1	Sex (male vs female)	0.183	0.183	10.299	.002
2	Stim trajectory (STG vs MTG; frontal excl	0.254	0.071	4.314	.044
3	Age (years)	0.257	0.003	0.171	.681
4	Stim hemisphere (R vs L)	0.280	0.022	1.332	.255
5	Stim amplitude (DB)	0.313	0.033	2.027	.162

~~Note. Each row shows the change produced by adding that predictor after all predictors above it were already in the model.~~

Unadjusted Sex Difference

Raw patient-level outcome by sex



Welch's test of the raw patient-level outcome gave $t(42.19) = 2.781$, $p = .008$, with males higher than females by 0.218 (95% CI [0.060, 0.377]). The adjusted primary model then evaluated sex first, followed by stimulation trajectory and the remaining covariates in that order.

Sensitivity Models

The sensitivity models show what happens when IED_freq and Memory_Z are brought into the analysis. The cost is lower N and, for Memory_Z, a narrower set of study cohorts. Within those reduced subsets, no additional requested predictor produced a clear signal.

Table 6
Comparison of fitted patient-level models

Model	N	Predictors used	Omitted in subset	R2	Adj R2	Model p
Primary Full-Sampl	48	Sex (male vs female); Stim trajec		0.313	0.231	.006
IED Frequency Se	41	Sex (male vs female); Stim trajec		0.286	0.160	.060
Memory_Z Sensiti	25	Sex (male vs female); Stim trajec	Stim amplitude (DB) (no va	0.137	-0.090	.698
Complete-Case Exp	10	Sex (male vs female); Stim trajectory	Stim amplitude (DB) (no va	0.000	-0.222	.999

Note. In the Memory_Z-only and complete-case subsets, stim_DB was automatically omitted because all available cases were at 0.5 DB, leaving no within-subset variation to estimate its effect.

IED Frequency Sensitivity Model

Sensitivity model adding IED frequency after the primary covariates. This costs additional patients because IED_freq is missing for part of the sample.

Table
IED Frequency Sensitivity Model coefficients

Term	B	SE	t	p	95% CI
Intercept	0.128	0.216	0.593	.557	[-0.310, 0.566]
Sex (male vs female)	0.089	0.102	0.875	.388	[-0.118, 0.297]
Stim trajectory (STG vs MTG; frontal	0.337	0.138	2.435	.020	[0.056, 0.618]
Age (years)	0.001	0.005	0.152	.880	[-0.009, 0.010]
Stim hemisphere (R vs L)	-0.044	0.090	-0.491	.626	[-0.226, 0.138]
Stim amplitude (DB)	-0.316	0.214	-1.477	.149	[-0.750, 0.119]
IED frequency (Likert)	-0.001	0.069	-0.007	.994	[-0.140, 0.139]

Note. Female, MTG trajectory, and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 41, R2 = 0.286, adjusted R2 = 0.160, F(6, 34) = 2.271, p = .060. Omitted predictors: None.

Table
IED Frequency Sensitivity Model sequential model-building steps

Step	Added predictor	Model R2	Delta R2	F change	p
1	Sex (male vs female)	0.115	0.115	5.073	.030
2	Stim trajectory (STG vs MTG; frontal exclus	0.222	0.107	5.214	.028
3	Age (years)	0.225	0.003	0.137	.714
4	Stim hemisphere (R vs L)	0.240	0.015	0.710	.405
5	Stim amplitude (DB)	0.286	0.046	2.273	.141
6	IED frequency (Likert)	0.286	0.000	0.000	.994

Note. Predictors are entered in the requested order for that model and delta R2 values are subset-specific.

Memory_Z Sensitivity Model

Sensitivity model adding pre-surgical baseline memory after the primary covariates. The available sample is restricted to the studies where Memory_Z exists.

Table
Memory_Z Sensitivity Model coefficients

Term	B	SE	t	p	95% CI
Intercept	-0.030	0.179	-0.167	.869	[-0.405, 0.345]
Sex (male vs female)	0.127	0.118	1.073	.297	[-0.120, 0.373]
Stim trajectory (STG vs MTG; frontal	0.148	0.232	0.639	.530	[-0.337, 0.633]
Age (years)	0.002	0.005	0.364	.720	[-0.009, 0.012]
Stim hemisphere (R vs L)	-0.048	0.137	-0.346	.733	[-0.335, 0.240]
Pre-surgical baseline memory (Memory_Z)	0.030	0.088	-0.342	.736	[-0.215, 0.154]

Note. Female, MTG trajectory, and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 25, R2 = 0.137, adjusted R2 = -0.090, F(5, 19) = 0.604, p = .698. Omitted predictors: Stim amplitude (DB) (no variation in analysis subset).

Table
Memory_Z Sensitivity Model sequential model-building steps

Step	Added predictor	Model R2	Delta R2	F change	p
1	Sex (male vs female)	0.097	0.097	2.482	.129
2	Stim trajectory (STG vs MTG; frontal exclus	0.125	0.028	0.693	.414
3	Age (years)	0.129	0.004	0.097	.758
4	Stim hemisphere (R vs L)	0.132	0.003	0.063	.804
5	Pre-surgical baseline memory (Memory_Z)	0.137	0.005	0.117	.736

Note. Predictors are entered in the requested order for that model and delta R2 values are subset-specific.

Complete-Case Exploratory Model

Exploratory complete-case model including all requested between-subject predictors. This is the most data-limited model and is shown mainly as a sensitivity check.

Table
Complete-Case Exploratory Model coefficients

Term	B	SE	t	p	95% CI
Intercept	0.038	0.261	0.145	.887	[-0.530, 0.606]
Sex (male vs female)	-0.000	0.168	-0.001	.999	[-0.367, 0.366]
Stim trajectory (STG vs MTG; frontal	0.283	0.304	0.930	.371	[-0.379, 0.945]
Age (years)	-0.001	0.009	-0.113	.912	[-0.020, 0.018]
Stim hemisphere (R vs L)	0.060	0.186	0.322	.753	[-0.346, 0.466]
IED frequency (Likert)	0.011	0.102	0.110	.914	[-0.210, 0.233]
Pre-surgical baseline memory (Memory_Z)	0.018	0.134	-0.136	.894	[-0.310, 0.273]

Note. Female, MTG trajectory, and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 19, R2 = 0.119, adjusted R2 = -0.322, F(6, 12) = 0.269, p = .941. Omitted predictors: Stim amplitude (DB) (no variation in analysis subset).

Table
Complete-Case Exploratory Model sequential model-building steps

Step	Added predictor	Model R2	Delta R2	F change	p
1	Sex (male vs female)	0.023	0.023	0.392	.540
2	Stim trajectory (STG vs MTG; frontal exclus	0.107	0.085	1.515	.236
3	Age (years)	0.107	0.000	0.005	.945
4	Stim hemisphere (R vs L)	0.117	0.010	0.154	.700
5	IED frequency (Likert)	0.117	0.000	0.004	.954
6	Pre-surgical baseline memory (Memory_Z)	0.119	0.001	0.018	.894

Note. Predictors are entered in the requested order for that model and delta R2 values are subset-specific.

Bottom Line

This file supports a between-subject regression question, not a within-subject MLM question. The outcome is already one number per patient, so there is no residual subject-level nesting to model.

In the full-sample adjusted model, sex was entered first and stimulation trajectory second. $B = 0.152$, 95% CI $[-0.019, 0.322]$, $p = .080$ for sex, and $B = 0.274$, 95% CI $[0.028, 0.521]$, $p = .030$ for STG versus MTG trajectory after adjustment.

Sequentially, sex produced model $R^2 = 0.183$ and adding trajectory increased R^2 by 0.071. Later additions were smaller and did not reverse the earlier pattern.

IED_freq did not contribute meaningfully in the sensitivity model when it was added after the primary covariates (step $p = .994$).

Memory_Z was only available for 31 patients from a restricted subset of studies, and its coefficient was not significant in that subset (step $p = .736$). Because all cases in that subset were at 0.5 DB, stim_DB could not be estimated there.

Recommended interpretation
If the substantive question is still what explains the high ICC from the original trial-level null model, the next step is to return to the trial-level dataset and add these patient-level covariates there as fixed effects in a mixed model. The present report instead shows which patient-level covariates relate to the collapsed patient summary outcome. Among the requested subject-level variables, sex and stimulation trajectory now enter the patient-level model first and should be interpreted together. The later covariates mainly test whether that earlier between-subject pattern survives adjustment rather than replacing it. Age, hemisphere, amplitude, IED_freq, and Memory_Z added comparatively modest information in these patient-level models.